

18.1 The homologous series of aldehydes and ketones

You briefly looked at aldehydes and ketones, the main classes of carbonyl compounds, in [Chapter 17](#).

IMPORTANT

Remember that:

- aldehydes can be formed from the oxidation of primary alcohols
- ketones can be formed from the oxidation of secondary alcohols (see [Section 18.2](#)).

In aldehydes, the carbon atom in the carbonyl group, $>\text{C}=\text{O}$, is bonded to a carbon atom and a hydrogen atom. In other words, the carbonyl group is positioned at the end of a carbon chain.

In ketones, the carbonyl group is attached to two other carbon atoms. Tables 18.1 and 18.2 give examples from the start of these homologous series.

Name	Structural formula
methanal	HCHO
ethanal	CH_3CHO
propanal	$\text{CH}_3\text{CH}_2\text{CHO}$
butanal	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$
pentanal	$\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$

Table 18.1: The names of aldehydes are derived from the name of the equivalent alkane, with the '-e' at the end of the name replaced by '-al'. Note that numbers are not needed when naming aldehydes, as the carbonyl group is always at the end of the carbon chain.

Name	Structural formula
propanone	CH_3COCH_3
butanone	$\text{CH}_3\text{COCH}_2\text{CH}_3$
pentan-2-one	$\text{CH}_3\text{COCH}_2\text{CH}_2\text{CH}_3$
pentan-3-one	$\text{CH}_3\text{CH}_2\text{COCH}_2\text{CH}_3$

Table 18.2: Ketones are named by replacing the '-e' at the end of the alkane with '-one'. However, in ketone molecules that are larger than butanone we also need to indicate the position of the carbonyl group.

Question

1 a Name the following compounds:

- $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CHO}$
- $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{COCH}_3$

b Draw the displayed formula of:

- methanal
- propanal

iii pentan-3-one.

c Draw the skeletal formula of the compounds listed in part a.

Now compare your answers to Question 1 with another learner and discuss any differences. Then check against the correct answers.

18.2 Preparation of aldehydes and ketones

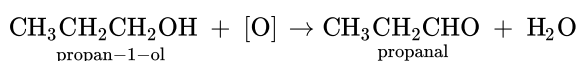
Preparing aldehydes

The general equation for the reaction in which an aldehyde is made from a primary alcohol is:

IMPORTANT

primary alcohol + oxygen atom from oxidising agent $\rightarrow$ aldehyde + water

For example:



The oxidising agent used is a solution of potassium dichromate(VI), which is orange, acidified with dilute sulfuric acid. To make the aldehyde, the primary alcohol is heated gently with acidified dichromate solution. The reaction mixture turns green as the orange dichromate ions, $\text{Cr}_2\text{O}_7^{2-}(\text{aq})$, are reduced to green $\text{Cr}^{3+}(\text{aq})$ ions.

PRACTICAL ACTIVITY 18.1

Oxidation of a primary alcohol

The acidified oxidising agent is added one drop at a time to the warm *primary* alcohol. The aldehyde made is distilled off as it forms in the reaction vessel. This method works because the aldehyde product will always have a lower boiling point than the alcohol reactant. If the aldehyde is not distilled off as soon as it is formed, further heating with acidified dichromate solution will oxidise the aldehyde produced to a carboxylic acid. The apparatus used to prepare a sample of ethanal is shown in Figure 18.2.

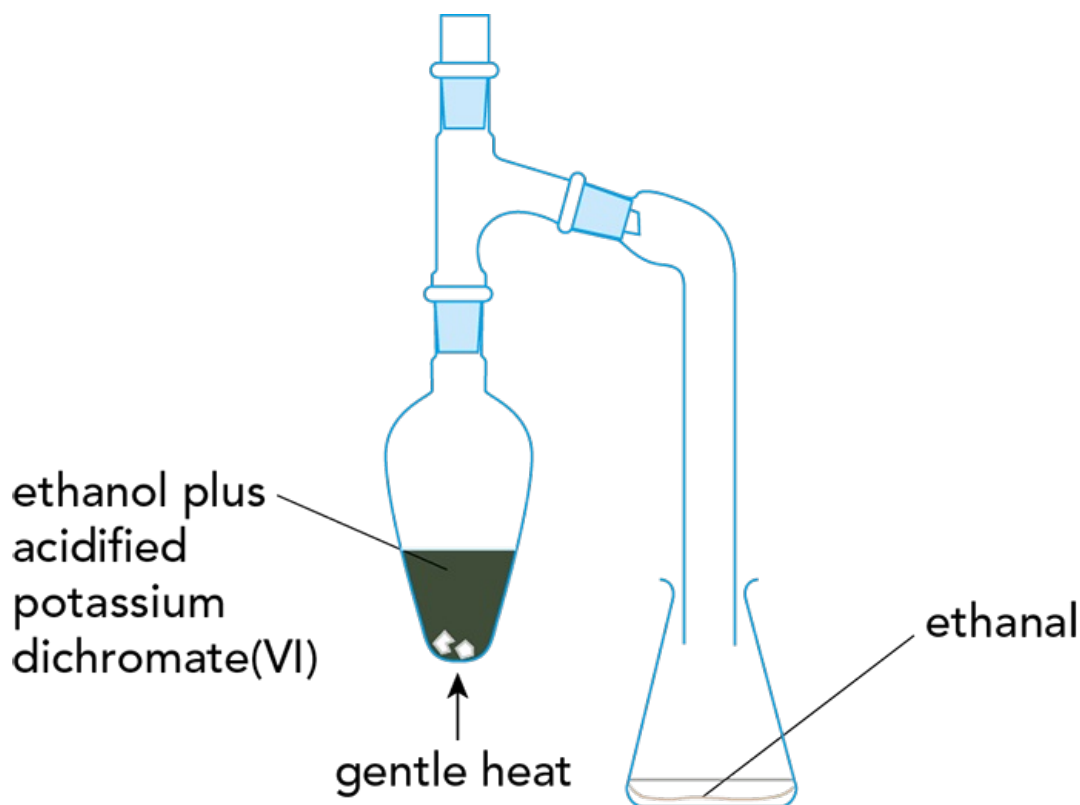


Figure 18.2: Distilling off and collecting the aldehyde, ethanal, formed in the mild oxidation of a primary alcohol, ethanol. The aqueous ethanal formed smells like rotting apples.

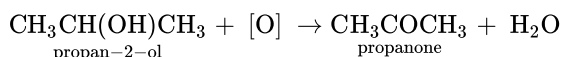
Preparing ketones

The general equation for making a ketone is:

IMPORTANT

secondary alcohol + oxygen atom from oxidising agent → ketone + water

For example:



Once again, the oxidising agent used is a solution of potassium dichromate(VI), acidified with dilute sulfuric acid.

PRACTICAL ACTIVITY 18.2

Oxidation of a secondary alcohol

To produce a ketone, the acidified oxidising agent must be heated with a *secondary* alcohol. The reaction mixture in the flask will change from orange to green.

Unlike aldehydes, the ketone formed cannot be further oxidised, even if you heat the reaction mixture under reflux and add excess oxidising agent. Therefore, we do not need to distil off the ketone product immediately.

Question

- 2 a i** Write a balanced equation for the oxidation of ethanol to ethanal, using [O] to represent an oxygen atom from the oxidising agent.
- ii** Give practical details to explain how you would use the reaction described in part **a i** to prepare and collect a sample of ethanal.
- b i** Write a balanced equation for the oxidation of butan-2-ol to butanone, using [O] to represent an oxygen atom from the oxidising agent.
- ii** What do you observe in the reaction vessel if the oxidising agent used in part **b i** is potassium dichromate(VI) solution, acidified with dilute sulfuric acid, and the reaction mixture is heated?
- iii** Discuss each option to complete the following statement with another learner, and agree on the correct answer:
- The butanone does not have to be distilled off as it forms in the reaction vessel because ...
- A** ... it cannot be oxidised to a carboxylic acid.
- B** ... it has a higher boiling point than butan-2-ol.
- C** ... it is easier to filter it off from the reaction mixture.
- D** ... it could be reduced back to butan-2-ol when heated.

18.3 Reduction of aldehydes and ketones

Chemical reduction of an aldehyde or ketone produces an alcohol:

IMPORTANT

aldehyde + reducing agent → primary alcohol

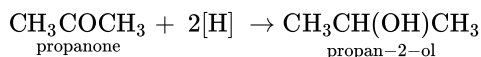
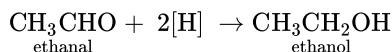
ketone + reducing agent → secondary alcohol

The reducing agent used is usually an aqueous alkaline solution of sodium tetrahydridoborate, NaBH_4 , or lithium tetrahydridoaluminate, LiAlH_4 , in dry ether.

The reduction reaction is carried out by either:

- warming the aldehyde or ketone with an aqueous alkaline solution of sodium tetrahydridoborate, or
- adding lithium tetrahydridoaluminate dissolved in a dry ether, such as diethyl ether, at room temperature. The organic solvent has to be dry because lithium tetrahydridoaluminate is a more powerful reducing agent than sodium tetrahydridoborate and reacts vigorously in water.

In the same way that we have used the symbol [O] in organic oxidation equations, we use the symbol [H] in reduction equations. [H] represents a hydrogen atom from the reducing agent. Look at the equations below used to summarise these reduction reactions:



Question

- 3 a** Write a balanced equation for the reaction that takes place when propanal is warmed with an aqueous alkaline solution of sodium tetrahydridoborate, using the symbol [H] to represent a hydrogen atom from the reducing agent.
- b** Name the product formed in the reduction reaction if pentan-3-one is added to lithium tetrahydridoaluminate in dry ether.

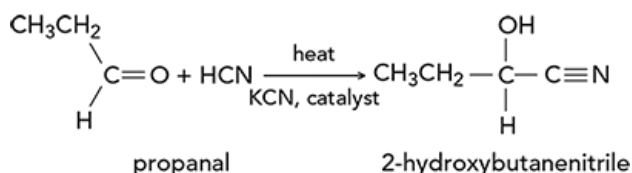
Compare your answers with those of another learner to arrive at two agreed answers.

18.4 Nucleophilic addition with HCN

The addition reactions we have met so far have involved electrophilic addition across the C=C double bond in alkene molecules (see [Section 15.3](#)).

Aldehydes and ketones also both undergo addition reactions with hydrogen cyanide, HCN. In these reactions, addition of HCN takes place across the C=O bond. However, the attack is by a nucleophile, not an electrophile.

We can show this using the **nucleophilic addition** reaction of the aldehyde propanal when heated with HCN. The HCN is generated *in situ* (in the reaction vessel) by the reaction of potassium cyanide, KCN, and dilute sulfuric acid.

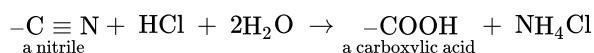


Note that a carbon atom has been added to the propanal molecule by the addition of the nitrile group ($-\text{C}\equiv\text{N}$). This is a useful reaction for chemists making new compounds as it increases the length of the carbon chain in the original aldehyde molecule by one carbon atom.

IMPORTANT

The nitrile group ($-\text{C}\equiv\text{N}$) can then be easily hydrolysed to a carboxylic acid.

You can carry out the hydrolysis of the nitrile group by refluxing with dilute hydrochloric acid:



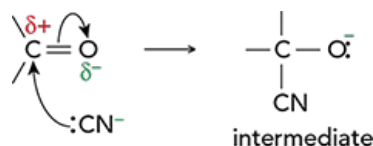
You can also hydrolyse the nitrile group by refluxing with dilute alkali, such as sodium hydroxide solution. In this reaction a sodium salt of the carboxylic acid, $-\text{COO}^-\text{Na}^+$, is formed. Therefore, if you want to produce the carboxylic acid, $-\text{COOH}$, the products are treated with a strong acid, such as dilute sulfuric acid.

Mechanism of nucleophilic addition

The carbonyl group, $>\text{C}=\text{O}$, in aldehydes and ketones is polarised due to the high electronegativity of the oxygen atom. The electrons in the C=O bond are drawn nearer to the O atom, giving it a partial negative charge, and leaving the C atom with a partial positive charge. This makes the carbonyl C atom open to attack by a nucleophile, such as the cyanide ion, CN^- .

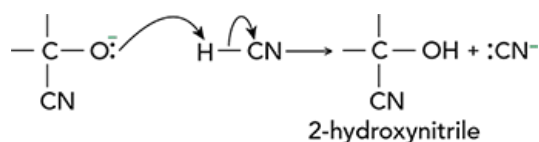
Note that the actual negative charge on a cyanide ion is on the carbon atom, not the nitrogen atom.

Step 1:



The negatively charged intermediate formed in the first step in the mechanism is highly reactive and quickly reacts with an aqueous H^+ ion. The hydrogen ion could be from HCN, from dilute acid or from water present in the reaction mixture. This forms the **2-hydroxynitrile** product.

Step 2:



All aldehydes and ketones form '2-hydroxynitriles' when they undergo nucleophilic addition. This is because the OH (hydroxyl) group will always be on the adjacent carbon atom to the nitrile group.

However, starting with the aldehyde methanal, HCHO , the hydroxynitrile formed in its nucleophilic addition with HCN would be called hydroxyethanenitrile. In this case there is no need to insert the '2' at the start of its name as the OH group could only possibly bond to the single carbon atom next to the nitrile group.

Question

4 Working with a partner, answer these questions together:

a Name the organic product that would be formed in the nucleophilic addition of HCN to:

i ethanal

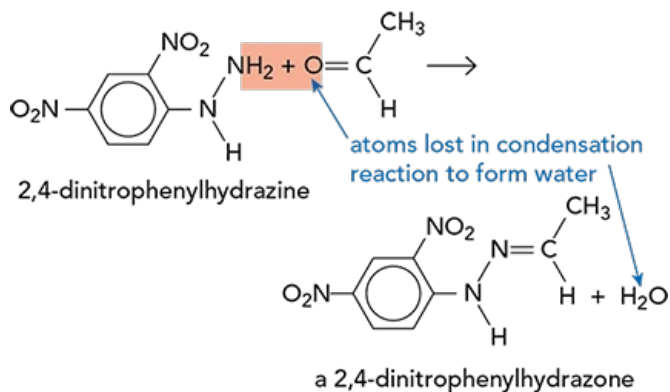
ii propanone.

b Use diagrams and curly arrows to describe the mechanism of the reaction in part **a i**.

18.5 Testing for aldehydes and ketones

Testing for the carbonyl group

The reaction of an aldehyde or ketone with 2,4-dinitrophenylhydrazine is an example of a *condensation reaction*. In a condensation reaction, two molecules join together and, in the process, eliminate a small molecule, in this case water. The equation for the reaction of ethanal with 2,4-DNPH is:



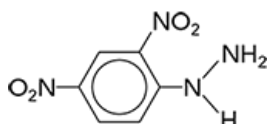
Other classes of organic compound that also contain the carbonyl group, $>\text{C}=\text{O}$ (such as carboxylic acids and esters), do not form precipitates with 2,4-DNPH.

PRACTICAL ACTIVITY 18.3

Testing with 2,4-DNPH

We can easily test for the presence of a carbonyl group in an aldehyde or ketone by adding a solution of 2,4-dinitrophenylhydrazine (often abbreviated to 2,4-DNPH). If an aldehyde or ketone is present, a *deep-orange precipitate* is formed (Figure 18.3).

The structure of 2,4-dinitrophenylhydrazine is:



The precipitate formed can be purified by recrystallisation and its melting point can be measured experimentally. The identity of the precipitated compound can then be found by referring to melting point data. From this, we can identify the specific aldehyde or ketone used in the test.

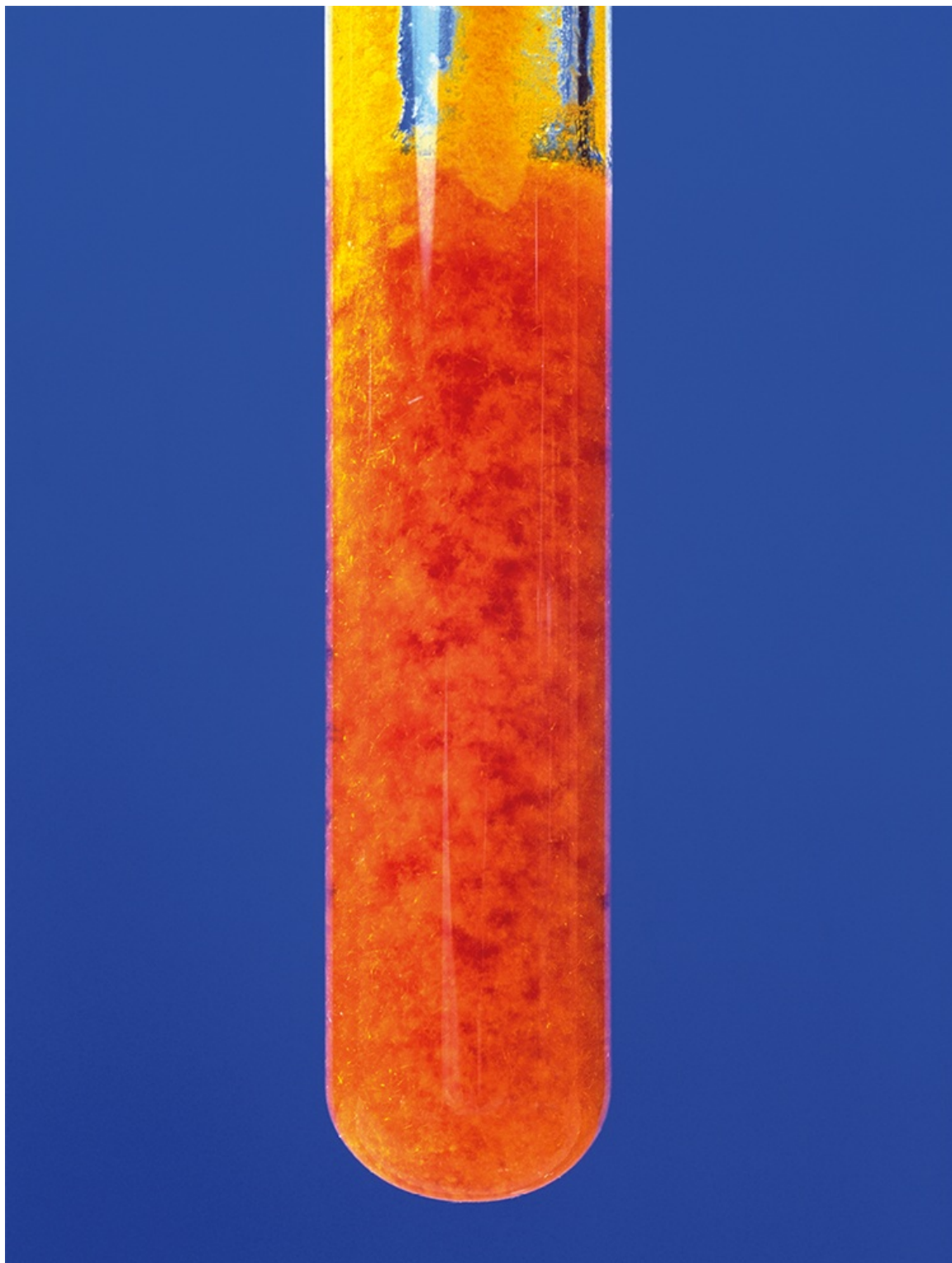


Figure 18.3: The orange precipitate formed from 2,4-DNPH in a test with propanone, a ketone.

Distinguishing between aldehydes and ketones

As we saw in [Section 18.2](#), aldehydes can be further oxidised to form carboxylic acids, but ketones cannot be oxidised easily. We can use this difference to distinguish between an aldehyde and a ketone in simple chemical tests. The two most common tests involve mild oxidation using Tollens' reagent or Fehling's solution.

practiCAL ACTIVITY 18.4

Testing with Tollens' reagent

Tollens' reagent is an aqueous solution of silver nitrate in excess ammonia solution, sometimes called *ammoniacal silver nitrate solution*. The silver ions, Ag^+ , in the solution act as a mild oxidising agent. When warmed, the Ag^+ ions will oxidise an aldehyde to form a carboxylate ion.

Under the alkaline conditions in the test, any carboxylic acid formed is immediately neutralised to the carboxylate ion, COO^- , as H^+ is removed from COOH and a salt is formed.

In the redox reaction with an aldehyde, the Ag^+ ions themselves are reduced to silver atoms. The silver atoms form a 'mirror' on the inside of the tube, giving a positive test for an aldehyde (Figure 18.4).

There will be *no change observed* when a ketone is warmed with Tollens' reagent as no redox reaction takes place. It remains a colourless mixture in the test-tube.



Figure 18.4: The 'before' and 'after' observations when Tollens' reagent is warmed with an aldehyde, such as ethanal, CH_3CHO .

PRACTICAL ACTIVITY 18.5

Testing with Fehling's solution

Fehling's solution is an alkaline solution containing copper(II) ions. When warmed with an aldehyde, the Cu^{2+} ions act as an oxidising agent. The aldehyde is oxidised to a carboxylate ion while the Cu^{2+} ions are reduced to Cu^+ ions. The clear blue Fehling's solution turns an opaque red / orange colour as a precipitate of copper(I) oxide forms throughout the solution (Figure 18.5).

As with Tollens' reagent, ketones are not oxidised in this test, so there is no change. The Fehling's solution remains blue when warmed.



Figure 18.5: 'The 'before' and 'after' observations when Fehling's solution is warmed with an aldehyde, such as ethanal, CH_3CHO .

Question

- 5 The melting points of the derivatives of the reaction between 2,4-DNPH and various aldehydes and ketones are shown in the table.

Product of reaction between 2,4-DNPH and...	Melting point / °C
ethanal	168
propanal	155
butanal	126
propanone	126
butanone	116

- a Describe your observations when each of the carbonyl compounds in the table is mixed with 2,4-DNPH.
- b A derivative was formed between 2,4-DNPH and an unknown carbonyl compound.

- i** The melting point of the derivative was 126 °C. Deduce what this result tells you.
 - ii** The unknown carbonyl compound formed an orange precipitate when warmed with Fehling's solution. Name the unknown compound.
 - iii** Describe and explain the different results obtained when the compound named in part **b ii** is warmed with Tollens' reagent in a test-tube and then the same test is performed on butanone.
- c** Write a half-equation to show silver ions acting as an oxidising agent in a positive test for an aldehyde.
- d** Write a half-equation to show copper(II) ions acting as an oxidising agent in a positive test for an aldehyde.

REFLECTION

At this point, join another learner to discuss and reflect on your progress so far in this chapter.

- 1** Do you know how to prepare aldehydes and ketones?
- 2** Can you write equations for their reduction and the oxidation of aldehydes?
- 3** What about explaining the mechanism of nucleophilic addition, as well as testing for aldehydes and ketones?
- 4** Discuss with a partner the thought processes you need to go through to construct the mechanism for the nucleophilic condensation reaction between ethanal and HCN.

18.6 Reactions to form tri-iodomethane

Tri-iodomethane (iodoform) forms as a yellow precipitate with methyl ketones, i.e. compounds containing the $\text{CH}_3\text{CO}-$ group (Figure 18.6). Note that ethanal, CH_3CHO , an aldehyde, also contains the $\text{CH}_3\text{CO}-$ group. Chemists use the appearance of the yellow precipitate as evidence of the $\text{CH}_3\text{CO}-$ group in an unknown compound.

The reagent used is an alkaline solution of iodine, which is warmed together with the substance being tested.



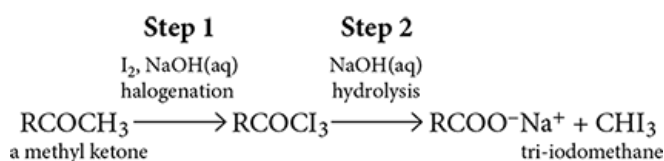
Figure 18.6: The yellow precipitate of tri-iodomethane forming.

The reaction involves two steps:

- 1 the carbonyl compound is halogenated: the three hydrogen atoms in the CH_3 methyl group are replaced by iodine atoms
- 2 the intermediate is hydrolysed to form the yellow precipitate of tri-iodomethane, CHI_3 (Figure 18.6).

When separated from the reaction mixture, the yellow crystals of tri-iodomethane can be positively identified by their melting point of 119°C .

Here, R is an alkyl group in a methyl ketone:



Testing for the $\text{CH}_3\text{CH}(\text{OH})-$ group

The tri-iodomethane test can also be used to identify the presence of a secondary alcohol, with the hydroxy

group (-OH) on the carbon atom next to a methyl group. This $\text{CH}_3\text{CH(OH)-}$ group is firstly oxidised by the alkaline iodine solution. This oxidation forms a methyl ketone, RCOCH_3 , which then reacts via the two steps shown at the bottom of the first column. The products of the reaction are the yellow tri-iodomethane precipitate, CHI_3 , and the sodium salt of a carboxylic acid, RCOO^-Na^+ .

Question

Work with a partner, if possible, to discuss your reasoning as you answer the questions below together:

- 6 a i** When propanone is warmed with alkaline iodine solution, a yellow precipitate is formed. Name and draw the displayed formula of the yellow precipitate.
- ii** Give the structural formulae of the organic products formed in both steps of the reaction in part **i**.
- b** Explain, naming any organic products formed, why ethanol gives a positive test (yellow precipitate) when warmed with alkaline iodine solution.
- c** Which of these compounds will give a yellow precipitate when treated with alkaline aqueous iodine?
- A** butanone
 - B** butanal
 - C** pentan-3-one
 - D** pentan-2-one
 - E** ethanal
 - F** methanol

At this point in the Coursebook you have finished the organic reactions needed for the AS Level course. You can be asked to devise a synthetic route (a series of reactions) to make a named product using any of the reactions in [Chapters 14 to 18](#). A useful summary of these reactions is given in Table 29.4 found in [Section 29.3](#). You can also try Question 8 at the end of this chapter as an example of devising synthetic routes.